FACTORIZATION THROUGH AFFINE SPACE OF RELATIVE DIMENSION R WITH UNRAMIFIED PROJECTION AT A POINT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and ends with the Nakayama step; the criterion for unramifiedness is cited but the construction of the factorization is entirely inside the counted span.

Editorial note. The counted spans carry 5 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-unramfied-over-affine is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- The module of differentials of a finite type morphism is finite and restricts to the fibre. Morphisms, Lemmas morphisms-lemma-finite-type-differentials and morphisms-lemma-base-change-differentials, invoked at proof lines 3488-3493. Supplies finiteness of the dimension r and the identification of the fibre of Omega_{X/S} with Omega_{X_s/s}.
- Tangent space as the dual of the fibre of the cotangent module. Same chapter, Lemma lemma-tangent-space-cotangent-space, invoked at proof line 3494. Supplies the identification of r with dim T_{X/S,x}.
- Criterion for unramifiedness by vanishing of the module of differentials. Algebra, Lemma algebra-lemma-unramified, invoked at proof lines 3508-3509. Supplies that Omega_{B/P,q} = 0 is what has to be shown.
- Nakayama's lemma for finite modules. Algebra, Lemma algebra-lemma-NAK, invoked at proof lines 3515-3516. Supplies the passage from the vanishing of the fibre of Omega_{B/P} at q to the vanishing of Omega_{B/P,q} itself.

Source reference. Original lemma 00DV: sources/stacks-chow/tags.txt, source lines 512–512. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 00UV: sources/stacks-chow/tags.txt, source lines 1125–1125. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0B2D: sources/stacks-varieties/pinned-original-source.tex, source lines 3014–3014. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01V0: sources/stacks-chow/tags.txt, source lines 2429–2429. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01V2: sources/stacks-chow/tags.txt, source lines 2431–2431. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $f : X \to S$ be a finite type morphism of affine schemes. Let $x \in X$ with image $s \in S$. Let*

$$r = \dim_{\kappa(x)} \Omega_{X/S,x} \otimes_{\mathcal{O}_{X,x}} \kappa(x) = \dim_{\kappa(x)} \Omega_{X_s/s,x} \otimes_{\mathcal{O}_{X_s,x}} \kappa(x) = \dim_{\kappa(x)} T_{X/S,x}$$

Then there exists a factorization

$$X \xrightarrow{\pi} \mathbf{A}^r_S \to S$$

of f such that π is unramified at x.

Proof. By Morphisms, Lemma 01V2 the first dimension is finite. The first equality follows as the restriction of $\Omega_{X/S}$ to the fibre is the module of differentials from Morphisms, Lemma 01V0. The last equality follows from Lemma 0B2D. Thus we see that the statement makes sense.

To prove the lemma write $S = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ and let $A \to B$ be the ring map corresponding to f. Let $\mathfrak{q} \subset B$ be the prime ideal corresponding to x. Choose a surjection of A-algebras $A[x_1, \ldots, x_t] \to B$. Since $\Omega_{B/A}$ is generated by $\mathrm{d}x_1, \ldots, \mathrm{d}x_t$ we see that their images in $\Omega_{X/S,x} \otimes_{\mathcal{O}_{X,x}} \kappa(x)$ generate this as a $\kappa(x)$-vector space. After renumbering we may assume that $\mathrm{d}x_1, \ldots, \mathrm{d}x_r$ map to a basis of $\Omega_{X/S,x} \otimes_{\mathcal{O}_{X,x}} \kappa(x)$. We claim that $P = A[x_1, \ldots, x_r] \to B$ is unramified at $\mathfrak{q}$. To see this it suffices to show that $\Omega_{B/P,\mathfrak{q}} = 0$ (Algebra, Lemma 00UV). Note that $\Omega_{B/P}$ is the quotient of $\Omega_{B/A}$ by the submodule generated by $\mathrm{d}x_1, \ldots, \mathrm{d}x_r$. Hence $\Omega_{B/P,\mathfrak{q}} \otimes_{B_\mathfrak{q}} \kappa(\mathfrak{q}) = 0$ by our choice of $x_1, \ldots, x_r$. By Nakayama's lemma, more precisely Algebra, Lemma 00DV part (2) which applies as $\Omega_{B/P}$ is finite (see reference above), we conclude that $\Omega_{B/P,\mathfrak{q}} = 0$. □